ATOMIC TRANSACTION

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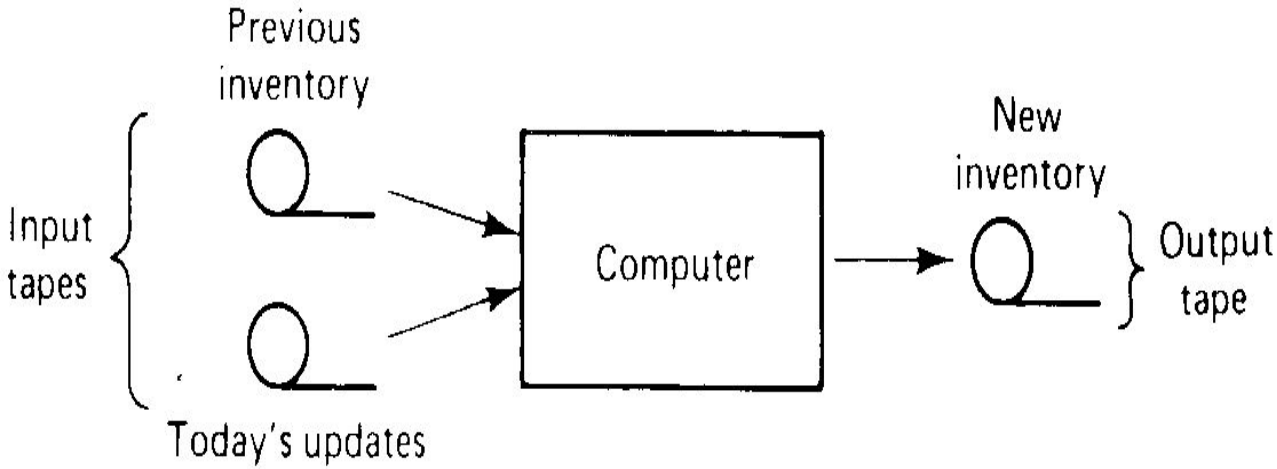
- In Distributed system , a transaction is an operation that involves multiple nodes .
- To maintain consistency and reliability these system must ensure that a transaction is atomic means it is treated as single , indivisible operation which is either commits all changes or none to the system.
- Use,Design and Implementation

ATOMIC TRANSACTION USE:



ATOMIC TRANSACTION USE:

1. Withdraw(amount, account1).
2. Deposit(amount, account2).

THE TRANSACTION MODEL

- Thee types of storage

1. Ordinary RAM
2. Disk storage
3. **Stable storage**: It can survive any thing except major calamities such as floods and Earthquacks.

Stable storage

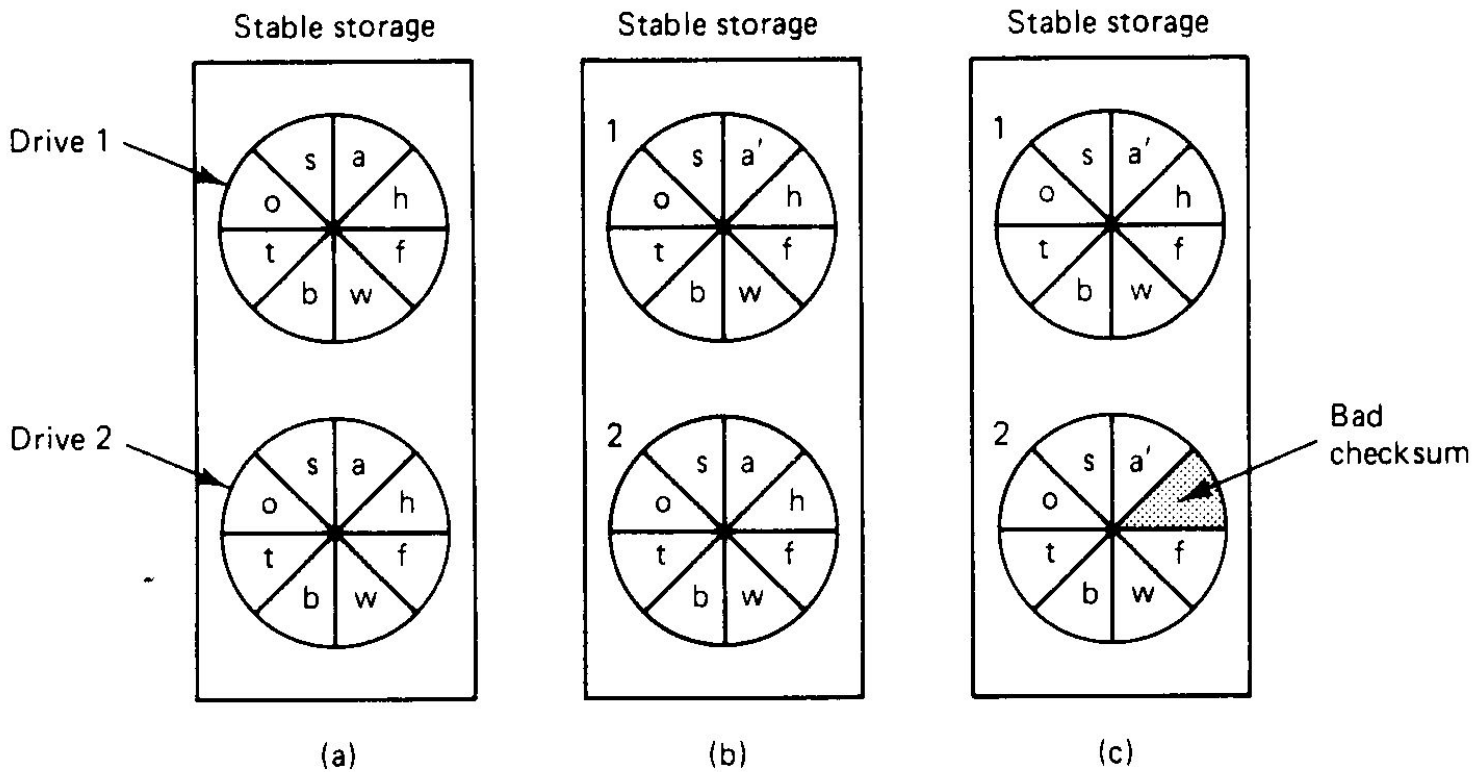


Fig. 3-15. (a) Stable storage. (b) Crash after drive 1 is updated. (c) Bad spot.

Transaction Primitives

Special primitives that must either be supplied by the operating system or language runtime system.

1. BEGIN_TRANSACTION: Mark the start of a transaction.
2. END_TRANSACTION: Terminate the transaction and try to commit.
3. ABORT_TRANSACTION: Kill the transaction; restore the old values.
4. READ: Read data from a file (or other object).
5. WRITE: Write data to a file (or other object).

Properties of Transaction

Transactions have four essential properties. Transactions are:

1. Atomic: To the outside world, the transaction happens indivisibly.
2. Consistent: The transaction does not violate system invariants.
3. Isolated: Concurrent transactions do not interfere with each other.
4. Durable: Once a transaction commits, the changes are permanent.

These properties are often referred to by their initial letters, **ACID**.

Nested Transaction

- Transaction may contain subtransaction,which is called nested transaction .
- Process->child process(fork)->subchild(fork)